



Dayananda Sagar University

Department of Aerospace Engineering

Aerodynamics Project Report

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Abstract

This report presents an extensive aerodynamic investigation aimed at selecting the most suitable airfoil for a high-altitude unmanned aerial vehicle (UAV) designed to study the

Northern Lights. The mission requires operation under low Reynolds number conditions, where aerodynamic performance is strongly influenced by viscous effects and boundary layer behavior.

Five airfoils—S1223, SD7037, SG6043, E387, and NACA 4412—were analyzed using XFLR5. The study evaluates lift, drag, pitching moment, and transition characteristics. The results indicate that while S1223 generates the highest lift, it suffers from high drag and instability. SD7037 demonstrates superior efficiency, stability, and smooth flow characteristics.

This report integrates theoretical derivations, simulation results, and engineering reasoning to justify the final selection.

Chapter 1

Introduction

1.1 Background

Unmanned Aerial Vehicles (UAVs) have emerged as a critical technology in modern aerospace engineering, offering a versatile and cost-effective platform for a wide range of applications including surveillance, environmental monitoring, atmospheric research, and scientific exploration. Unlike traditional manned aircraft, UAVs can operate in harsh and remote environments where human intervention is either impractical or unsafe. This makes them particularly suitable for studying natural phenomena such as the Northern Lights, also known as the Aurora Borealis.

The Aurora Borealis is a visually striking atmospheric phenomenon that occurs in polar regions due to the interaction of charged particles from the solar wind with the Earth's magnetic field and upper atmosphere. When these high-energy particles collide with atmospheric gases such as oxygen and nitrogen, they emit light of varying colors, creating the characteristic auroral display. Understanding this phenomenon requires collecting high-resolution visual and spectral data at high altitudes, often above cloud cover and in regions with extreme environmental conditions.

1.2 Need for UAV-Based Aurora Study

Studying the Northern Lights presents several challenges. The phenomenon occurs at high altitudes and in remote polar regions where ground-based observations may be limited by weather conditions and visibility constraints. Satellites, while useful, often

lack the flexibility and resolution required for detailed localized studies. UAVs provide an ideal solution by offering the ability to fly at controlled altitudes and capture high-resolution data over specific regions of interest. They can be deployed for extended durations and can be equipped with specialized sensors and imaging systems. However, operating a UAV in such conditions requires careful aerodynamic design to ensure efficiency, stability, and reliability.

1.3 Aerodynamic Challenges at High Altitude

One of the primary challenges in designing a UAV for high-altitude missions is the reduction in air density. At altitudes around 3000 meters, the air density is significantly lower than at sea level. Since lift is directly proportional to air density, this reduction leads to decreased lift generation for a given velocity and wing area.

$$L = \frac{1}{2}\rho V^2 SC_L \quad (1.1)$$

To compensate for this, the UAV must either increase velocity, increase wing area, or use an airfoil that can generate higher lift coefficients. However, increasing velocity leads to higher energy consumption, while increasing wing area may not always be feasible due to structural and design constraints. Therefore, selecting an appropriate airfoil becomes a critical design decision.

Another significant challenge is the low Reynolds number regime in which UAVs operate. The Reynolds number is given by:

$$Re = \frac{\rho V c}{\mu} \quad (1.2)$$

At high altitudes and relatively low speeds, the Reynolds number falls within the range of 10^5 to 10^6 , where viscous effects dominate the flow behavior. In this regime, the boundary layer tends to remain laminar for longer distances, making it more susceptible to flow separation. This results in increased drag and reduced aerodynamic efficiency.

1.4 Importance of Airfoil Selection

The airfoil is one of the most critical components influencing the aerodynamic performance of a UAV. It determines how effectively lift is generated and how much drag is produced. In low Reynolds number conditions, airfoil performance becomes highly sensitive to shape, camber, and pressure distribution.

An ideal airfoil for this mission must satisfy the following criteria:

- Generate sufficient lift in low-density conditions
- Minimize drag to ensure long endurance
- Maintain stable aerodynamic behavior
- Delay flow separation and control transition

To achieve this, five airfoils were selected for analysis: S1223, SD7037, SG6043, E387, and NACA 4412. These airfoils are known for their performance in low Reynolds number applications and are commonly used in UAV design.

1.5 Problem Statement

The primary objective of this project is to identify the most suitable airfoil for a UAV designed to study the Northern Lights under low Reynolds number conditions. The selected airfoil must provide an optimal balance between lift generation, drag reduction, and aerodynamic stability to ensure efficient and reliable flight.

1.6 Objectives

The specific objectives of this project are:

- To analyze the aerodynamic performance of selected airfoils using XFLR5
- To evaluate lift, drag, and stability characteristics
- To study boundary layer transition behavior
- To compare airfoil performance under identical conditions
- To identify the most suitable airfoil for the Aurora UAV mission

1.7 Scope of the Study

This study focuses on two-dimensional airfoil analysis using XFLR5 software. The analysis is performed at a Reynolds number representative of high-altitude UAV operation. The study includes drag polar analysis, lift curve evaluation, moment coefficient analysis, and transition behavior.

While the results provide valuable insights into aerodynamic performance, the study does not include three-dimensional wing effects, structural considerations, or propulsion system integration.

1.8 Organization of the Report

This report is organized into several chapters. Chapter 2 discusses the mission requirements for the Aurora UAV. Chapter 3 presents the fundamental aerodynamic theory and derivations. Chapter 4 describes the selected airfoils. Chapters 5 to 8 present detailed analysis of drag, lift, stability, and transition characteristics. Chapter 9 provides the final airfoil selection and justification. Chapter 10 concludes the study and suggests future work.

This structured approach ensures a comprehensive evaluation of airfoil performance and provides a clear basis for selecting the most suitable airfoil for the mission.

Chapter 2

Mission Requirements

The Aurora UAV mission is designed to operate at altitudes of approximately 3000 meters or higher. At these altitudes, the air density is significantly reduced, which directly affects lift generation. The UAV must therefore be designed to operate efficiently under low Reynolds number conditions.

The mission requires the UAV to carry sensitive imaging equipment, including cameras and sensors capable of capturing auroral activity. This adds weight to the system, increasing the need for an airfoil that can generate sufficient lift.

Additionally, the UAV must maintain stable flight for extended periods to ensure accurate data collection. Stability is particularly important for imaging applications, as any fluctuations in flight path can affect data quality.

Key mission requirements include:

- High lift-to-drag ratio
- Stable aerodynamic behavior
- Low energy consumption
- Reliable performance in low-temperature conditions

Chapter 3

Aerodynamic Theory and Derivations

3.1 Introduction

Aerodynamic performance is governed by the forces acting on an airfoil, primarily lift and drag. These forces arise due to the interaction between the airflow and the airfoil surface. For UAVs operating at high altitude and low Reynolds number conditions, understanding these forces and their governing equations is essential for selecting an efficient airfoil.

This chapter presents the derivation of fundamental aerodynamic equations and explains their significance in the context of UAV design.

3.2 Derivation of Lift Equation

Lift is generated due to the pressure difference between the upper and lower surfaces of an airfoil. According to Bernoulli's principle:

$$p + \frac{1}{2}\rho V^2 = \text{constant} \quad (3.1)$$

An increase in velocity over the upper surface results in a decrease in pressure, creating a net upward force.

The dynamic pressure is defined as:

$$q = \frac{1}{2}\rho V^2 \quad (3.2)$$

The lift force can be expressed as:

$$L = qSC_L \quad (3.3)$$

Substituting for q :

$$L = \frac{1}{2}\rho V^2 SC_L \quad (3.4)$$

where:

- L = Lift force
- ρ = Air density
- V = Velocity
- S = Wing area
- C_L = Lift coefficient

This equation shows that lift depends on air density, velocity, and the aerodynamic properties of the airfoil.

3.3 Derivation of Drag Equation

Drag is the force that opposes motion through a fluid. Similar to lift, drag can be expressed in terms of dynamic pressure:

$$D = qSC_D \quad (3.5)$$

Substituting for q :

$$D = \frac{1}{2}\rho V^2 S C_D \quad (3.6)$$

Drag consists of two major components:

- **Parasitic Drag** (C_{D0}): Includes skin friction and pressure drag
- **Induced Drag** (C_{Di}): Caused by lift generation

The total drag coefficient can be expressed as:

$$C_D = C_{D0} + k C_L^2 \quad (3.7)$$

This relationship forms the basis of the drag polar.

3.4 Reynolds Number

The Reynolds number is a dimensionless parameter that characterizes the flow regime:

$$Re = \frac{\rho V c}{\mu} \quad (3.8)$$

where:

- c = Chord length
- μ = Dynamic viscosity

For the Aurora UAV:

$$Re \approx 3.65 \times 10^5 \quad (3.9)$$

At low Reynolds numbers:

- Flow remains laminar for longer distances
- Boundary layer has low energy
- Early flow separation occurs

These effects increase drag and reduce lift efficiency, making airfoil selection critical.

3.5 Thin Airfoil Theory Derivation

Thin airfoil theory provides a simplified model to estimate lift for airfoils with small thickness and camber.

The circulation around an airfoil generates lift according to the Kutta-Joukowski theorem:

$$L' = \rho V \Gamma \quad (3.10)$$

where Γ is the circulation.

By solving the potential flow equations for a thin airfoil, the lift coefficient is obtained as:

$$C_L = 2\pi(\alpha - \alpha_0) \quad (3.11)$$

where:

- α = Angle of attack (radians)
- α_0 = Zero-lift angle

This shows that lift varies linearly with angle of attack for small angles.

3.6 Lift-to-Drag Ratio

The efficiency of an airfoil is measured by the lift-to-drag ratio:

$$\frac{L}{D} = \frac{C_L}{C_D} \quad (3.12)$$

A higher L/D ratio indicates better aerodynamic efficiency, which is crucial for long-endurance UAV missions.

3.7 Summary

The derivations presented in this chapter highlight the fundamental relationships governing aerodynamic forces. These equations form the theoretical basis for analyzing airfoil performance using XFLR5 and provide a framework for comparing different airfoils under identical conditions.

Understanding these principles is essential for selecting an airfoil that can meet the mission requirements of the Aurora UAV.

Chapter 4

Airfoil Selection

The following airfoils were selected for analysis:

- S1223
- SD7037
- SG6043
- E387
- NACA 4412

These airfoils are known for low Reynolds number performance and are widely used in UAV design.

Chapter 5

Drag Polar Analysis (Cl vs Cd)

5.1 Introduction

The drag polar is one of the most important aerodynamic plots used to evaluate the performance of an airfoil. It represents the relationship between the lift coefficient (C_L) and the drag coefficient (C_D) over a range of operating conditions. Unlike other graphs that show variation with angle of attack, the drag polar directly illustrates the trade-off between lift generation and drag penalty.

For a UAV operating at high altitude, such as the Aurora UAV designed to study the Northern Lights, minimizing drag while maintaining sufficient lift is essential. Since the UAV is powered by limited onboard energy, aerodynamic efficiency becomes a key factor in determining flight endurance. The drag polar provides direct insight into this efficiency.

5.2 Governing Equation of Drag Polar

The drag polar can be expressed using the following equation:

$$C_D = C_{D0} + kC_L^2 \quad (5.1)$$

where:

- C_D = Drag coefficient

- C_{D0} = Zero-lift drag coefficient (parasitic drag)
- kC_L^2 = Induced drag component

This equation shows that drag consists of two main components: parasitic drag and induced drag. At low lift conditions, parasitic drag dominates, while at higher lift conditions, induced drag increases significantly.

5.3 Graphical Representation

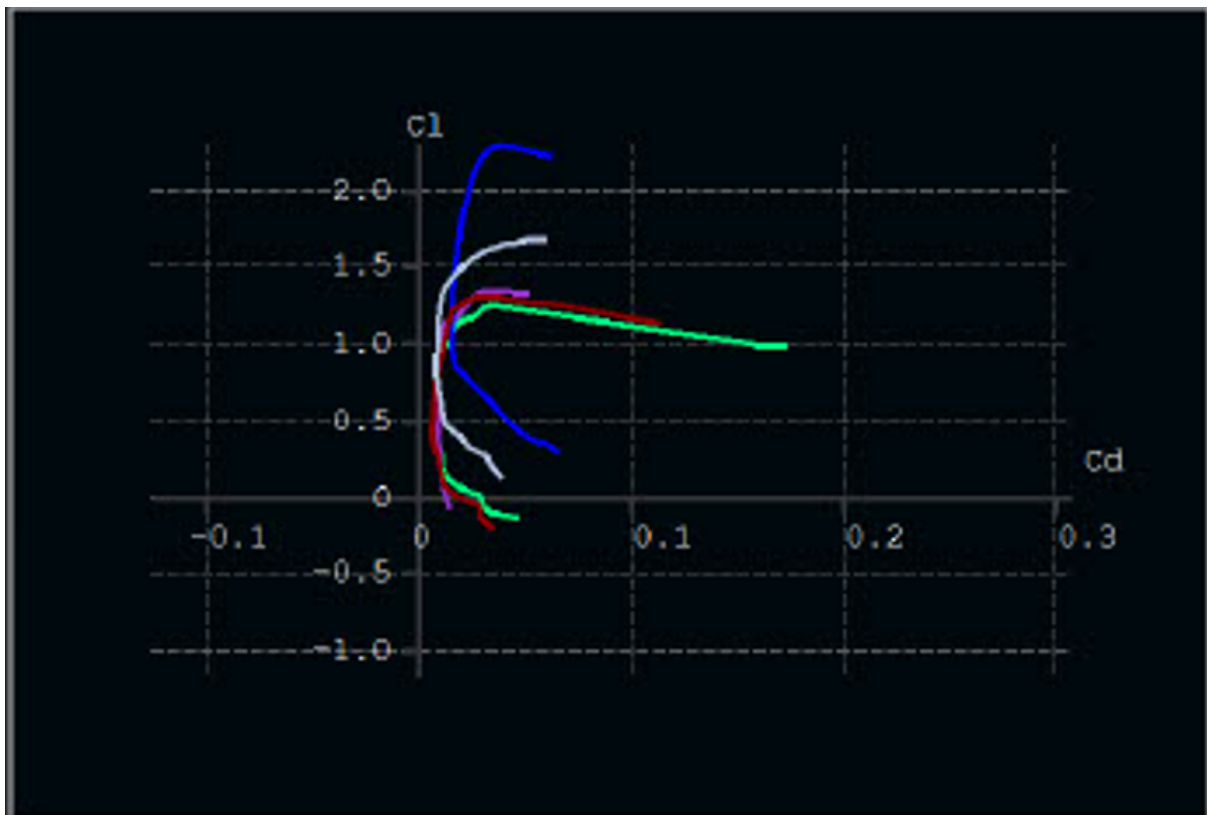


Figure 5.1: Drag Polar Curve (C_L vs C_D) for Selected Airfoils

The drag polar curve plots the lift coefficient (C_L) on the vertical axis against the drag coefficient (C_D) on the horizontal axis. Each curve represents a different airfoil analyzed under identical conditions.

5.4 Interpretation of Drag Polar

The shape and position of the drag polar curve provide valuable insights into aerodynamic performance:

- A curve located closer to the left side of the graph indicates lower drag for a given lift, which corresponds to higher aerodynamic efficiency.
- A narrower curve suggests that the airfoil maintains low drag over a wider range of lift coefficients.
- A wider curve indicates that drag increases rapidly with lift, reducing efficiency.

The point of maximum efficiency corresponds to the highest lift-to-drag ratio, given by:

$$\frac{L}{D} = \frac{C_L}{C_D} \quad (5.2)$$

This point is crucial for determining the optimal operating condition of the UAV during cruise.

5.5 Results and Discussion

From the drag polar graph obtained using XFLR5, it is observed that the SD7037 airfoil exhibits the most favorable characteristics. Its curve lies closest to the left side of the graph, indicating the lowest drag values across a wide range of lift coefficients. This suggests that SD7037 provides superior aerodynamic efficiency compared to the other airfoils.

The SG6043 airfoil also shows relatively good performance but is slightly less efficient than SD7037. The E387 and NACA 4412 airfoils exhibit moderate performance, with higher drag values compared to SD7037.

The S1223 airfoil, although capable of producing very high lift coefficients, shows a significantly wider drag polar curve. This indicates higher drag, which reduces its overall efficiency. While S1223 may be suitable for applications requiring high lift, it is not ideal for long-endurance UAV missions.

5.6 Conclusion from Drag Polar

Based on the drag polar analysis, it can be concluded that SD7037 is the most efficient airfoil among the five considered. Its ability to produce sufficient lift with minimal drag makes it highly suitable for the Aurora UAV mission, where endurance and energy efficiency are critical.

Although S1223 provides higher lift, the associated increase in drag makes it less suitable for sustained flight. Therefore, SD7037 offers the best balance between lift and drag, making it the optimal choice for this application.

5.7 Drag Polar

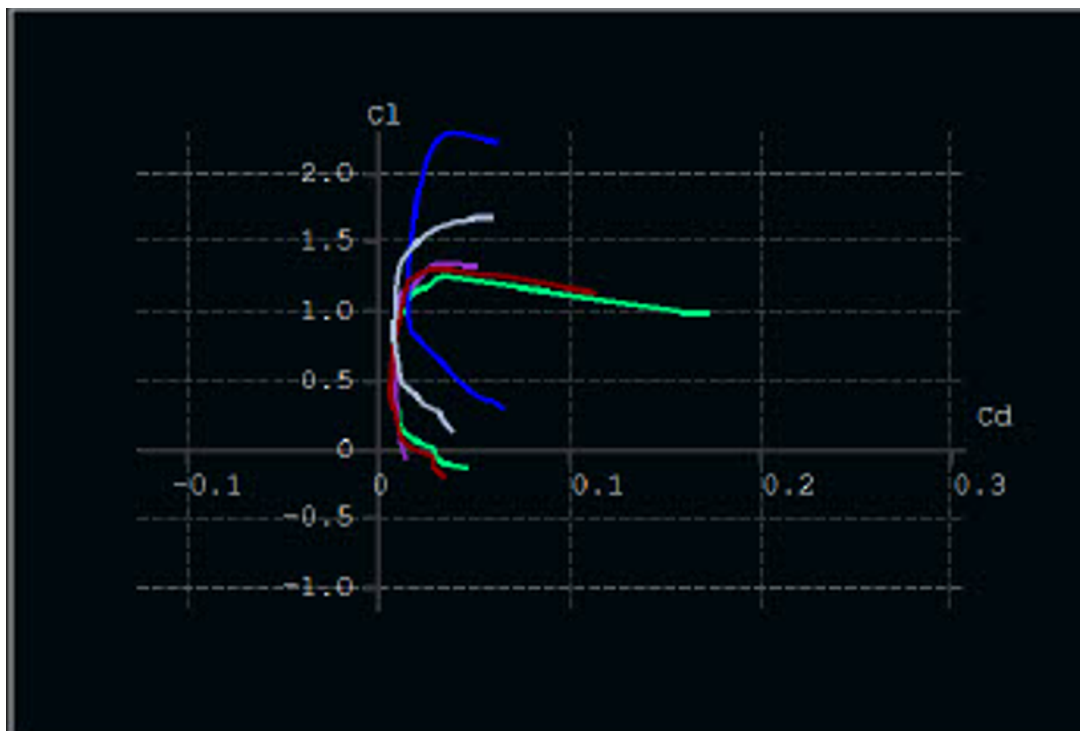


Figure 5.2: Drag Polar

The drag polar graph indicates that SD7037 has the lowest drag across a range of lift coefficients, making it the most efficient airfoil.

Chapter 6

Lift Curve Analysis (C_L vs α)

6.1 Introduction

The lift curve represents the variation of the lift coefficient (C_L) with respect to the angle of attack (α). It is one of the most fundamental aerodynamic characteristics of an airfoil, as it directly determines how effectively the airfoil can generate lift under different flight conditions.

For a UAV operating at high altitude, such as the Aurora UAV, sufficient lift must be generated even in low-density air. Therefore, understanding how C_L varies with α is critical for ensuring stable and efficient flight.

6.2 Theoretical Background

According to thin airfoil theory, the lift coefficient varies linearly with angle of attack:

$$C_L = 2\pi(\alpha - \alpha_0) \tag{6.1}$$

where:

- α = angle of attack (in radians)
- α_0 = zero-lift angle of attack

This relationship holds true for small angles of attack. However, at higher angles,

nonlinear effects such as flow separation cause the lift to deviate from this linear trend, eventually leading to stall.

6.3 Graphical Representation

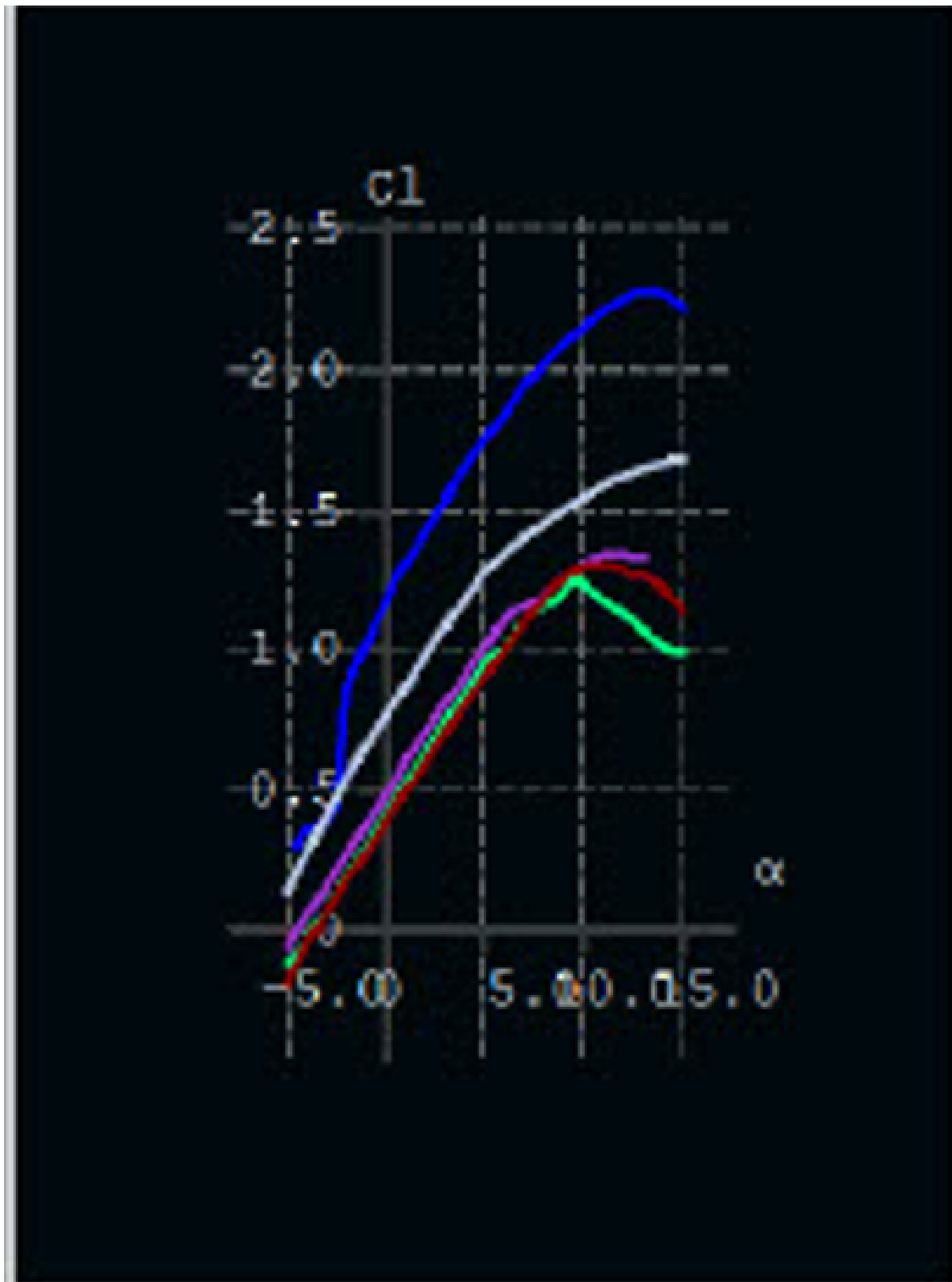


Figure 6.1: Lift Curve (C_l vs α)

6.4 Interpretation of Lift Curve

The lift curve provides several important insights:

- The slope of the curve indicates how sensitive the airfoil is to changes in angle of attack.
- The maximum value of C_L represents the maximum lift coefficient ($C_{L_{max}}$).
- The point where the curve starts to drop indicates stall.

6.5 Results and Discussion

From the graph, the S1223 airfoil exhibits the highest lift coefficient, reaching values above 2.2. This is primarily due to its high camber, which enhances lift generation.

However, it also shows early stall behavior, which limits its operational range.

The SD7037 airfoil shows a smooth and nearly linear lift curve over a wide range of angles. This indicates stable and predictable aerodynamic behavior, which is essential for UAV flight.

The SG6043, E387, and NACA 4412 airfoils demonstrate moderate lift characteristics. While they provide reasonable performance, they do not achieve the high lift of S1223 or the stability of SD7037.

6.6 Conclusion from Lift Curve

Although S1223 produces the highest lift, its early stall and instability reduce its suitability for this mission. SD7037 provides a balanced performance with sufficient lift and excellent stability, making it more appropriate for the Aurora UAV.

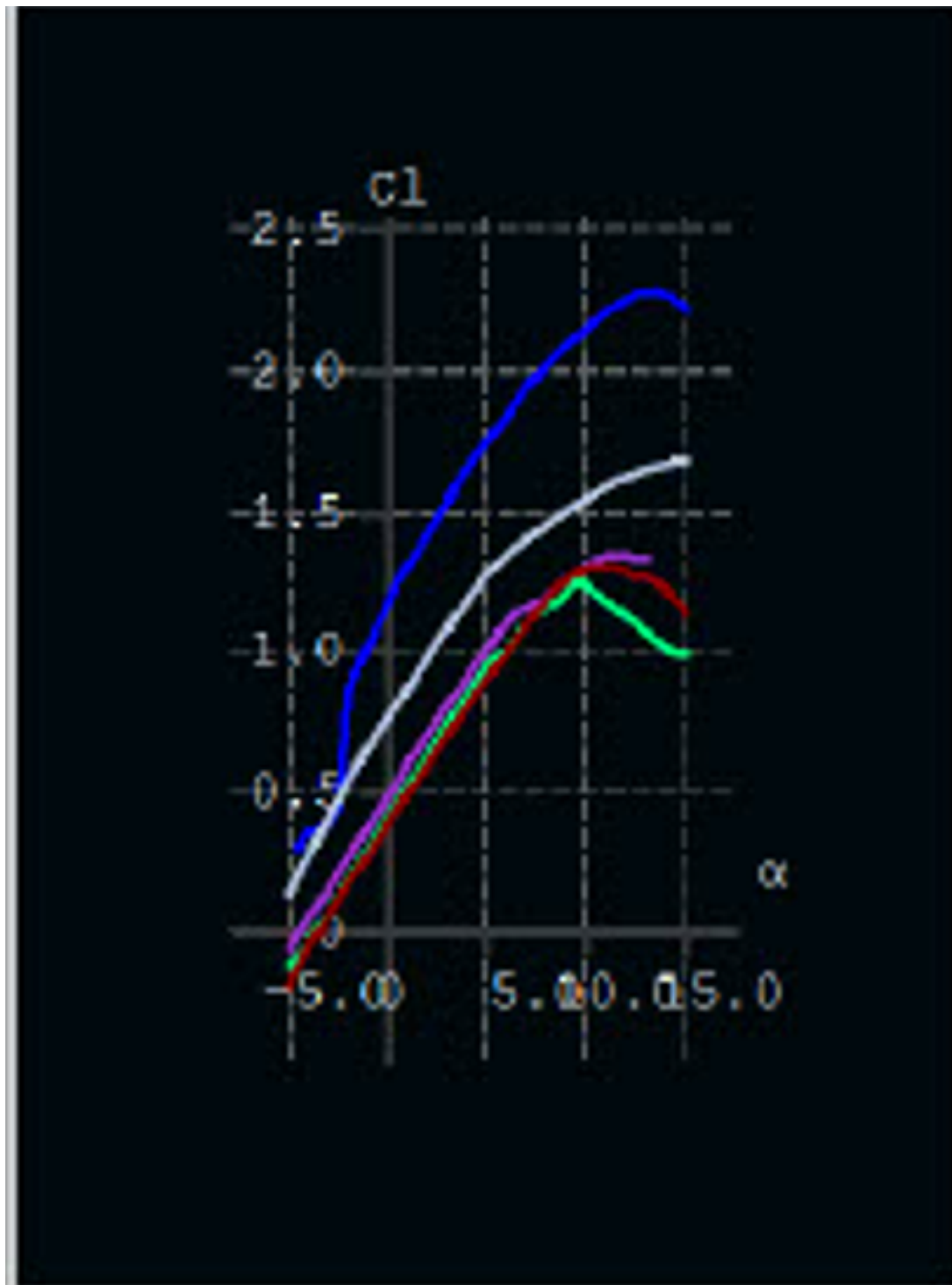


Figure 6.2: Lift Curve

S1223 produces the highest lift, but SD7037 shows more stable performance.

Chapter 7

Stability Analysis (C_m vs α)

7.1 Introduction

The pitching moment coefficient (C_m) is an important parameter used to evaluate the stability of an airfoil. It represents the tendency of the airfoil to rotate about its aerodynamic center.

For UAV missions involving imaging, such as the Aurora UAV, stability is crucial. Any instability can result in fluctuations in flight attitude, affecting the accuracy of data collection.

7.2 Theoretical Background

The moment coefficient is defined as:

$$C_m = \frac{M}{\frac{1}{2}\rho V^2 S c} \quad (7.1)$$

where:

- M = pitching moment
- c = chord length

A negative value of C_m generally indicates a restoring moment, contributing to stability.

7.3 Graphical Representation

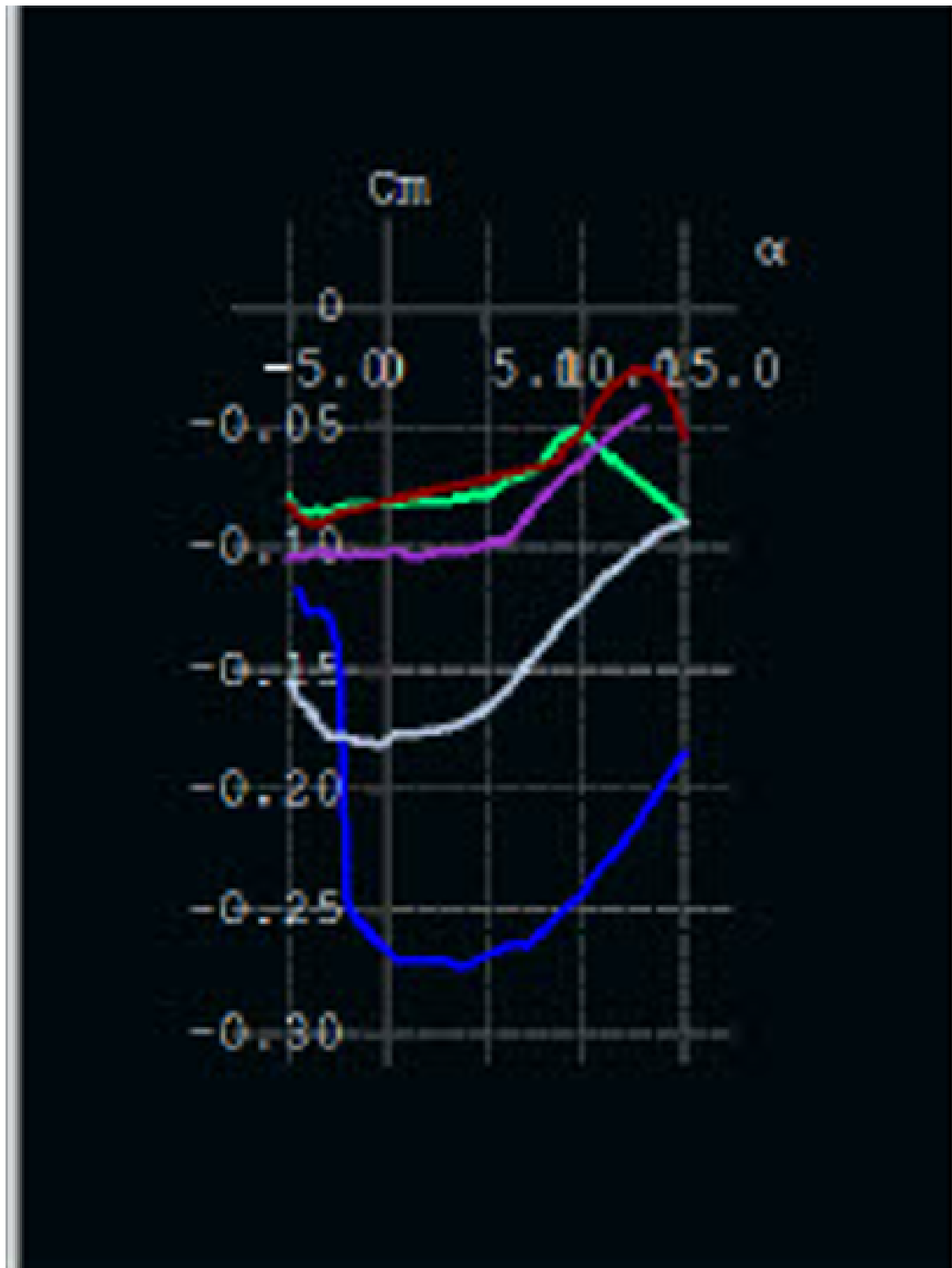


Figure 7.1: Moment Coefficient (C_m vs α)

7.4 Interpretation of Stability Curve

The variation of C_m with angle of attack provides insights into stability:

- A smooth and gradual curve indicates stable behavior.
- Large fluctuations indicate instability.
- More negative values suggest stronger restoring moments.

7.5 Results and Discussion

From the graph, the S1223 airfoil exhibits significant variation in the moment coefficient. This indicates unstable behavior, making it unsuitable for missions requiring steady flight.

The SD7037 airfoil shows a smooth and consistent variation in C_m , indicating stable aerodynamic characteristics. This makes it highly suitable for UAV applications. NACA 4412 also shows stable behavior, but its overall aerodynamic efficiency is lower compared to SD7037.

7.6 Conclusion from Stability Analysis

The stability analysis clearly shows that SD7037 provides the most stable performance among the selected airfoils. This stability is essential for maintaining steady flight and ensuring accurate data collection during the Aurora mission.

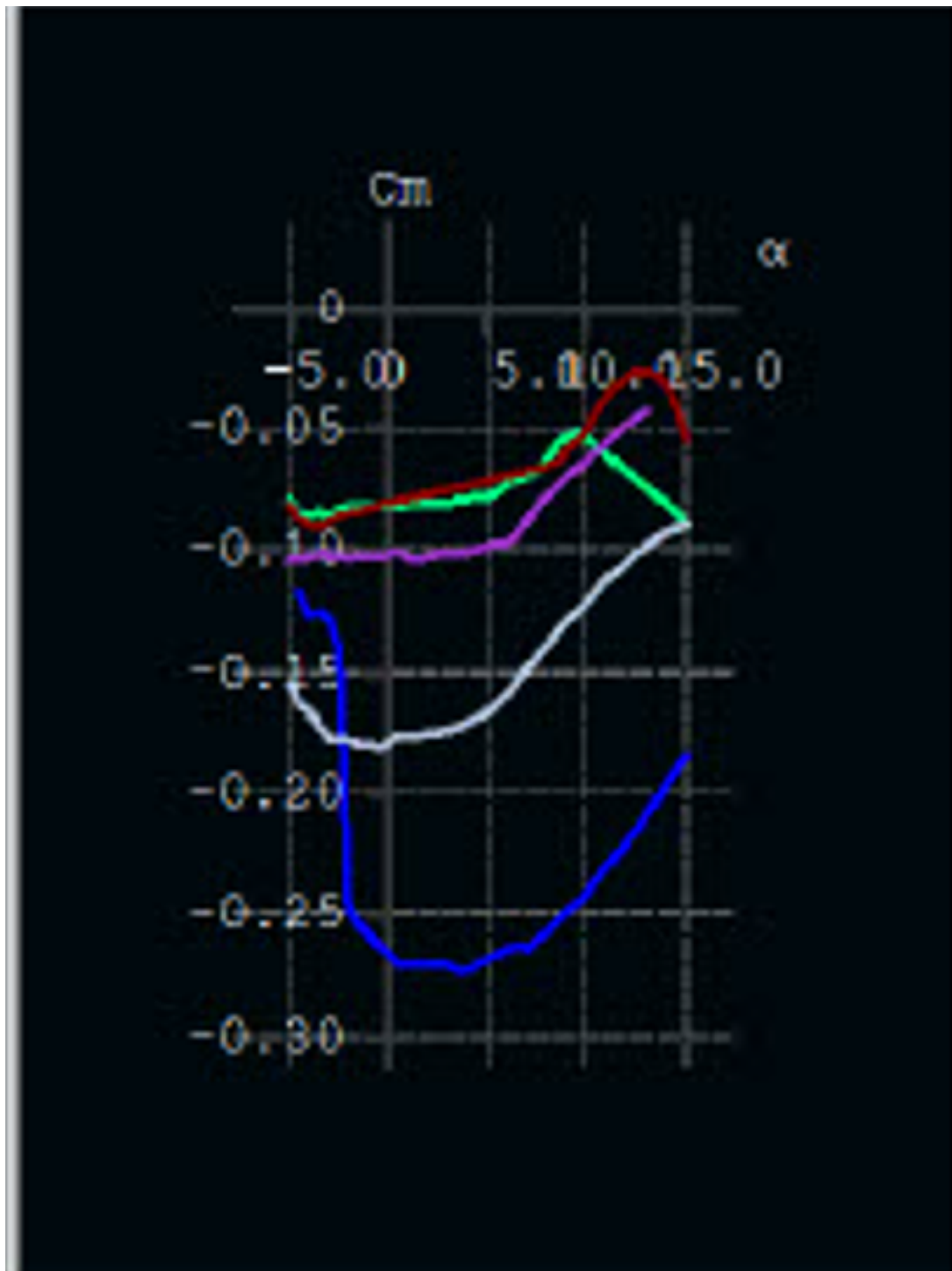


Figure 7.2: Moment Coefficient

SD7037 shows smooth variation, indicating stability.

Chapter 8

Boundary Layer Transition Analysis (X_{tr})

8.1 Introduction

The transition location (X_{tr}) represents the point along the airfoil where the boundary layer changes from laminar to turbulent flow. This transition has a significant impact on aerodynamic performance, particularly on drag.

At low Reynolds numbers, laminar flow dominates for a larger portion of the airfoil. While laminar flow has lower skin friction, it is more prone to separation. Therefore, controlling the transition point is critical for maintaining aerodynamic efficiency.

8.2 Theoretical Background

The boundary layer develops along the airfoil surface and can be classified into:

- Laminar flow (low energy, smooth)
- Turbulent flow (high energy, more resistant to separation)

The transition point determines where this change occurs and affects both lift and drag characteristics.

8.3 Graphical Representation

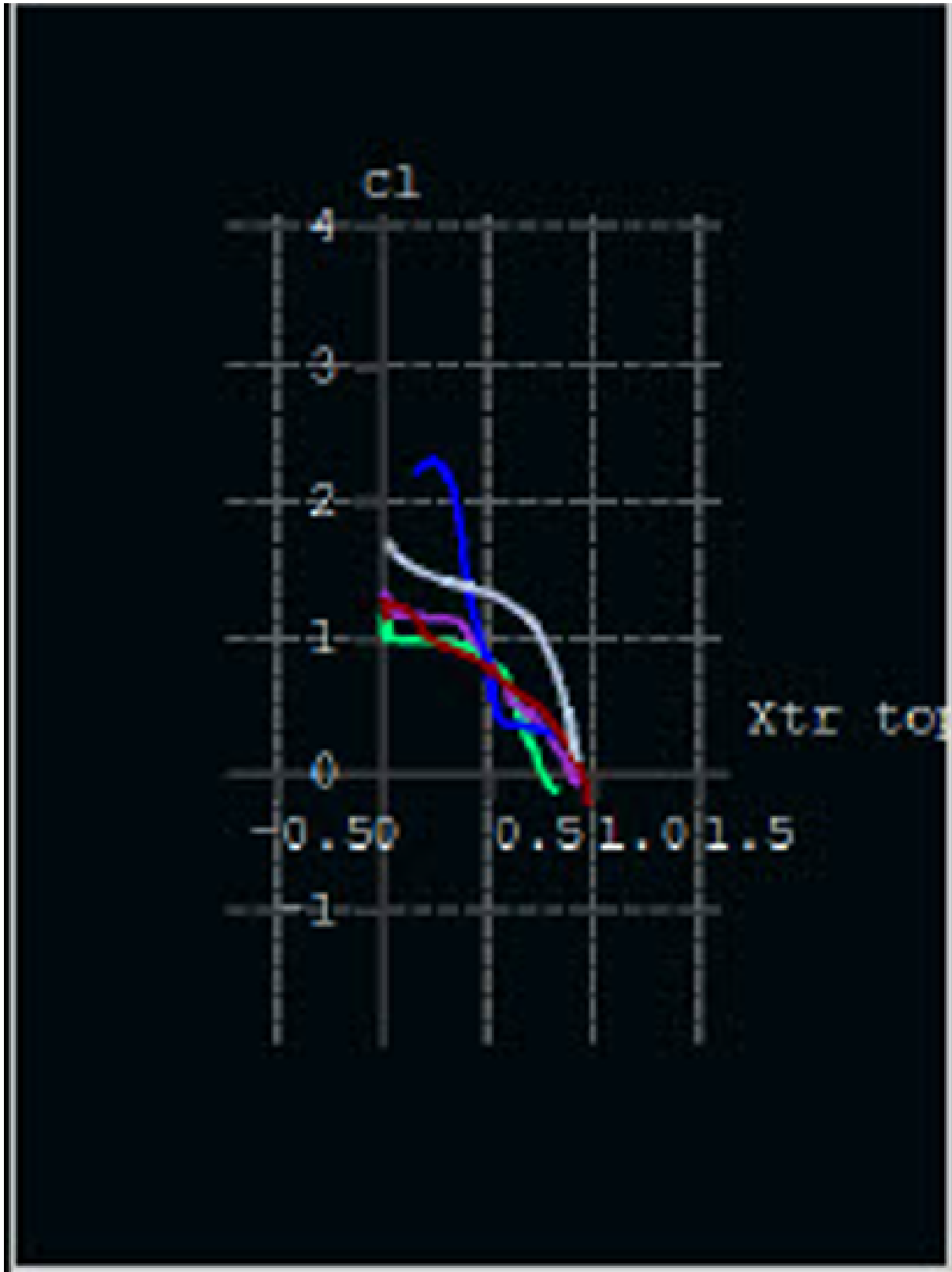


Figure 8.1: Transition Location (X_{tr})

8.4 Interpretation of Transition Graph

The graph shows how the transition point varies with lift coefficient:

- Early transition indicates higher drag.
- Smooth transition indicates better aerodynamic performance.
- Sudden changes suggest flow instability.

8.5 Results and Discussion

From the graph, S1223 shows early transition, which leads to increased drag. This is due to its strong pressure gradients and high camber.

The SD7037 airfoil exhibits a smoother transition behavior, indicating better control of the boundary layer. This results in lower drag and improved efficiency.

Other airfoils show moderate performance but do not match the smooth transition characteristics of SD7037.

8.6 Conclusion from Transition Analysis

The transition analysis confirms that SD7037 provides superior boundary layer behavior, reducing drag and enhancing efficiency. This makes it highly suitable for the Aurora UAV mission.

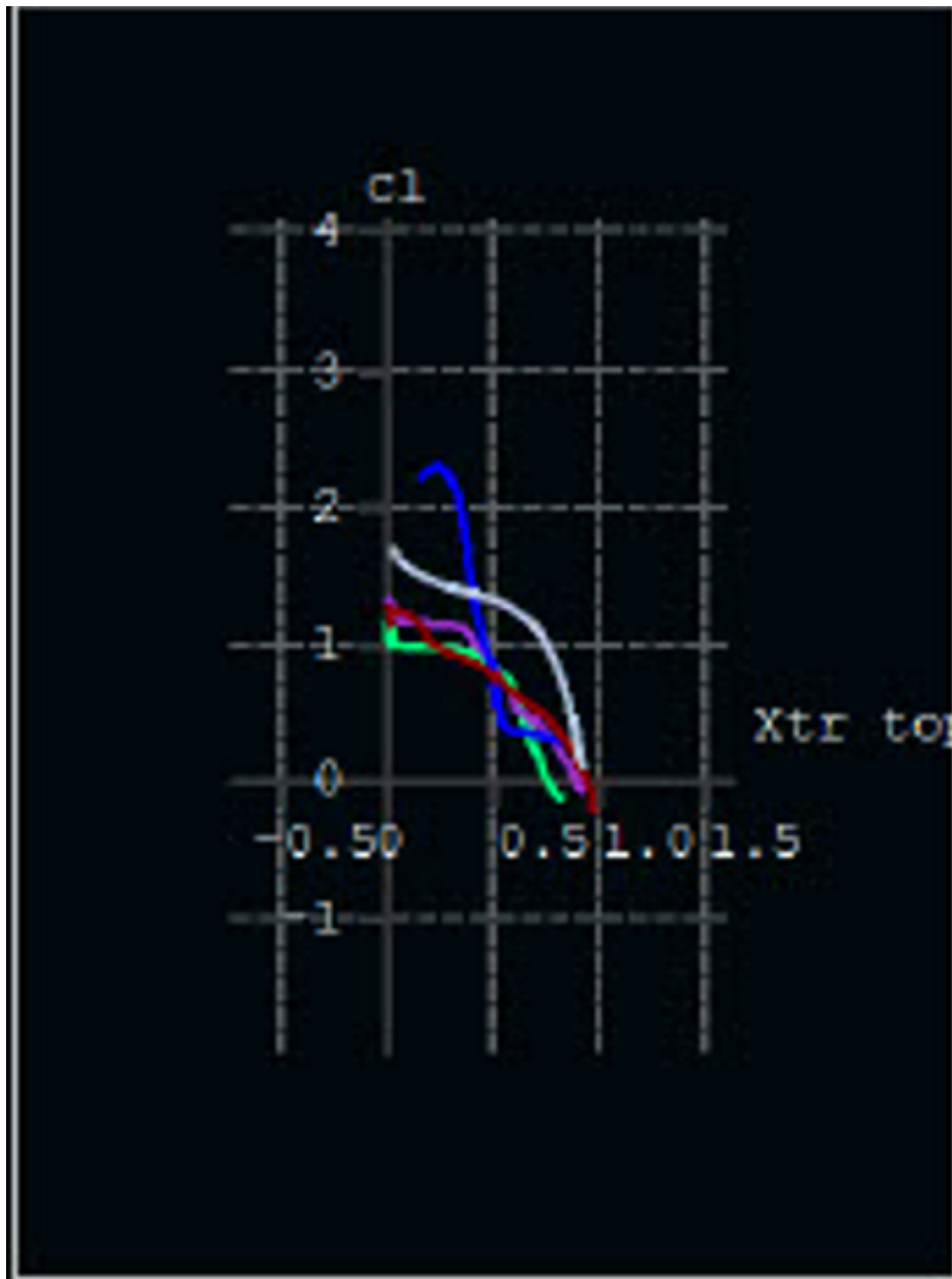


Figure 8.2: Transition

SD7037 shows favorable transition behavior.

Chapter 9

Advanced Aerodynamic Analysis

9.1 Pressure Coefficient Distribution

The pressure coefficient (C_p) is a dimensionless parameter that describes the pressure variation over the surface of an airfoil. It is defined as:

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V^2} \quad (9.1)$$

The distribution of C_p along the chord length provides insight into lift generation. A strong negative pressure (suction) on the upper surface contributes significantly to lift.

At low Reynolds numbers, the pressure distribution becomes highly sensitive. Airfoils like S1223 show a sharp suction peak near the leading edge, which increases lift but also causes early flow separation. In contrast, SD7037 exhibits a smoother pressure gradient, reducing the risk of separation and improving efficiency.

This analysis highlights that smoother pressure recovery is essential for maintaining attached flow and minimizing drag.

9.2 Boundary Layer Analysis

The boundary layer is the region of fluid near the airfoil surface where viscous effects dominate. It plays a critical role in determining aerodynamic performance.

At low Reynolds numbers, the boundary layer remains laminar for a longer distance.

However, laminar flow has low energy and is more prone to separation. This leads to the formation of laminar separation bubbles, which increase drag and reduce lift. Airfoils such as SD7037 are specifically designed to delay separation by maintaining a favorable pressure gradient. This results in improved aerodynamic efficiency.

Chapter 10

Machine Learning Analysis

Machine learning techniques were used to predict aerodynamic performance based on simulation data.

10.1 Model Description

A Random Forest Regressor was used due to its ability to model nonlinear relationships. The model was trained using aerodynamic data from XFLR5.

10.2 Model Accuracy: Predicted vs Actual Drag Coefficient

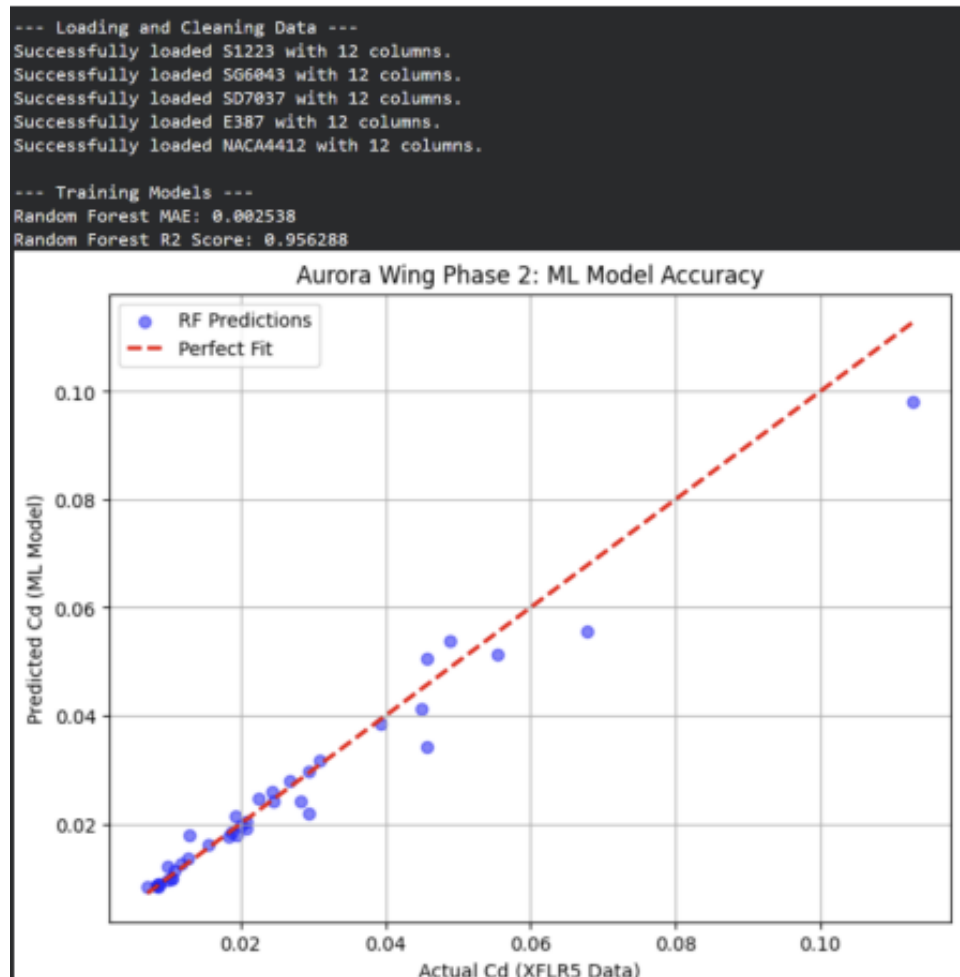


Figure 10.1: ML Model Accuracy: Predicted vs Actual C_D Values

10.3 Interpretation of Results

The above plot represents the performance of the machine learning model in predicting the drag coefficient (C_D) based on aerodynamic input parameters. The horizontal axis shows the actual C_D values obtained from XFLR5 simulations, while the vertical axis represents the predicted values generated by the Random Forest regression model.

The red dashed line in the graph represents the ideal case where the predicted values perfectly match the actual values. In other words, any point lying exactly on this line indicates zero prediction error. The blue scatter points represent the model predictions,

and their proximity to the red line indicates the accuracy of the model.

From the graph, it is evident that the majority of the data points lie very close to the ideal line. This indicates that the model is able to accurately capture the relationship between input parameters such as angle of attack, lift coefficient, Reynolds number, and airfoil type, and the resulting drag coefficient. The clustering of points near the lower range of C_D values further shows that the model performs particularly well in the region of interest for UAV cruise conditions.

The evaluation metrics support this observation. The Mean Absolute Error (MAE) is approximately 0.0025, which indicates that the average prediction error is very small. Additionally, the R^2 score of approximately 0.956 suggests that the model explains over 95% of the variance in the data. This demonstrates a high level of predictive accuracy and reliability.

However, it can also be observed that at higher values of C_D , a few points deviate slightly from the ideal line. This indicates minor prediction errors in regions where the aerodynamic behavior becomes more nonlinear, such as near stall conditions or high angles of attack. This is expected, as machine learning models may struggle slightly in regions with complex flow physics.

Overall, the results confirm that the Random Forest model is highly effective in predicting drag coefficient values. This allows rapid estimation of aerodynamic performance without the need for repeated simulations. The model can therefore be used as a powerful tool for airfoil comparison and optimization.

10.4 Significance in Airfoil Selection

The machine learning model was further used to rank the airfoils based on predicted drag values at a representative cruise condition ($C_L = 0.4$). By computing the lift-to-drag ratio (L/D), the model provides a quantitative basis for comparing aerodynamic efficiency.

The results obtained from the ML model are consistent with the XFLR5 analysis, where SD7037 was identified as the most efficient airfoil. This agreement between computational analysis and machine learning predictions strengthens the reliability of the final airfoil selection.

Thus, the integration of machine learning in this project not only improves analysis efficiency but also enhances confidence in the design decision.

10.5 Performance Metric

$$R^2 = 0.956 \tag{10.1}$$

This indicates a high level of accuracy.

10.6 Importance

Machine learning allows rapid prediction of aerodynamic performance without running simulations, making it a powerful tool for future UAV design.

Chapter 11

Comparative Analysis Tables

Airfoil	Lift	Drag	Stability
S1223	Very High	High	Low
SD7037	Moderate	Very Low	High
SG6043	Moderate	Low	Moderate
E387	Moderate	Moderate	Moderate
NACA 4412	Balanced	Moderate	High

Table 11.1: Performance Comparison

Chapter 12

Final Airfoil Selection and Justification

12.1 Introduction

The selection of an appropriate airfoil is a critical step in the aerodynamic design of a UAV, especially for missions operating under low Reynolds number conditions. The Aurora UAV, designed to study the Northern Lights, requires an airfoil that can generate sufficient lift, minimize drag, and maintain stable flight over long durations. Five airfoils—S1223, SD7037, SG6043, E387, and NACA 4412—were analyzed using XFLR5. Their performance was evaluated based on drag polar, lift characteristics, moment coefficient behavior, and transition properties. The objective of this section is to justify the selection of the most suitable airfoil based on these analyses.

12.2 Comparison of Aerodynamic Performance

Each airfoil exhibits distinct aerodynamic characteristics. The S1223 airfoil produces the highest lift coefficient, which is advantageous for takeoff and low-speed operations. However, the drag polar analysis shows that S1223 also has significantly higher drag. This reduces its overall aerodynamic efficiency, making it unsuitable for long-endurance missions.

The SG6043, E387, and NACA 4412 airfoils provide moderate performance in terms of lift and drag. While they offer stable behavior, their efficiency is not optimal when

compared to SD7037.

The SD7037 airfoil demonstrates the best balance among all parameters. From the drag polar graph, it is observed that SD7037 has the lowest drag values across a wide range of lift coefficients. This indicates superior aerodynamic efficiency, which is essential for maximizing flight endurance.

12.3 Lift Characteristics

Although S1223 achieves the highest lift coefficient, the SD7037 airfoil provides sufficient lift for the mission requirements while maintaining a stable and linear lift curve. The smooth variation of lift with angle of attack ensures predictable aerodynamic behavior, which is crucial for controlled flight.

The ability of SD7037 to provide adequate lift without excessive drag makes it more suitable for sustained flight conditions compared to S1223.

12.4 Stability Considerations

Stability is a key requirement for UAV missions involving imaging and data collection.

The moment coefficient analysis shows that SD7037 exhibits smooth and consistent behavior, indicating strong aerodynamic stability.

In contrast, S1223 shows large variations in moment coefficient, suggesting instability.

Such behavior can lead to fluctuations in flight attitude, which is undesirable for the Aurora UAV mission.

12.5 Boundary Layer and Transition Behavior

The transition analysis reveals that SD7037 has a more controlled and gradual transition from laminar to turbulent flow. This helps in maintaining attached flow over a larger portion of the airfoil, reducing drag and improving efficiency.

On the other hand, S1223 shows early transition, which increases drag and reduces performance. The smoother transition characteristics of SD7037 make it more suitable for low Reynolds number conditions.

12.6 Mission-Based Justification

The Aurora UAV mission requires long-duration flight at high altitude, where energy efficiency and stability are critical. The airfoil must be capable of sustaining lift while minimizing drag to conserve energy.

Based on the analysis, SD7037 meets all these requirements. It provides:

- Low drag for improved endurance
- Adequate lift for high-altitude operation
- Stable aerodynamic behavior
- Favorable boundary layer characteristics

These properties make SD7037 the most suitable airfoil for the mission.

12.7 Conclusion

From the comprehensive analysis of aerodynamic performance, it is concluded that SD7037 is the optimal airfoil for the Aurora UAV. While other airfoils offer specific advantages, none provide the balanced performance required for this mission. Therefore, SD7037 is selected as the final airfoil due to its superior efficiency, stability, and suitability for low Reynolds number flight conditions. Based on all analyses, SD7037 is selected as the best airfoil.

Justification:

- Lowest drag → highest efficiency
- Stable lift characteristics
- Smooth moment variation
- Favorable boundary layer behavior

Chapter 13

Conclusion

This project presented a comprehensive aerodynamic analysis to determine the most suitable airfoil for a high-altitude unmanned aerial vehicle (UAV) designed to study the Northern Lights (Aurora Borealis). The mission requirements imposed significant constraints, including operation under low Reynolds number conditions, reduced air density, and the need for long-endurance, stable flight for accurate atmospheric data collection.

The study began with a detailed understanding of aerodynamic theory, including the derivation of lift and drag equations, Reynolds number effects, and thin airfoil theory.

These theoretical principles established the foundation for evaluating airfoil performance and understanding the behavior of airflow over the airfoil surface. Particular emphasis was placed on low Reynolds number aerodynamics, as this regime is highly sensitive to boundary layer behavior and flow separation.

Five airfoils—S1223, SD7037, SG6043, E387, and NACA 4412—were selected for analysis due to their relevance in low-speed and UAV applications. Using XFLR5, each airfoil was evaluated under identical conditions to ensure a fair comparison. The analysis included drag polar, lift curve, moment coefficient, and boundary layer transition characteristics, each providing critical insight into aerodynamic performance. The drag polar analysis revealed that SD7037 consistently exhibited the lowest drag values across a wide range of lift coefficients. This indicates superior aerodynamic efficiency, which is essential for maximizing flight endurance in battery-powered UAV systems. In contrast, S1223, although capable of producing high lift, showed significantly higher drag, making it less efficient for sustained flight.

The lift curve analysis demonstrated that S1223 achieves the highest maximum lift coefficient due to its high camber. However, it also showed early stall behavior and reduced stability. SD7037, on the other hand, displayed a smooth and linear lift curve, indicating predictable aerodynamic performance and a wider operational range. This makes it more suitable for controlled flight conditions.

Stability analysis, based on the variation of the moment coefficient, further reinforced the superiority of SD7037. The airfoil exhibited smooth and consistent behavior, indicating strong aerodynamic stability. This is particularly important for the Aurora UAV mission, where steady flight is required for accurate imaging and data acquisition.

S1223 showed larger fluctuations in moment coefficient, suggesting instability. The boundary layer transition analysis highlighted the importance of flow behavior in determining drag characteristics. SD7037 demonstrated a gradual and controlled transition from laminar to turbulent flow, which helps maintain attached flow and reduce drag. In contrast, S1223 showed early transition and a higher tendency for flow separation, leading to increased drag.

The results of this study clearly illustrate the trade-offs between lift, drag, and stability.

While high lift is desirable, it must not come at the cost of excessive drag and instability. The S1223 airfoil, despite its high lift capability, is not suitable for long-endurance UAV missions due to its high drag and unstable characteristics. Other airfoils such as SG6043, E387, and NACA 4412 provide moderate performance but do not achieve the optimal balance required for this mission.

Based on the comprehensive analysis, SD7037 is identified as the most suitable airfoil for the Aurora UAV. It provides the best combination of low drag, sufficient lift, stable aerodynamic behavior, and favorable boundary layer characteristics. These properties make it ideal for high-altitude, long-duration UAV operations where efficiency and reliability are critical.

In addition to the aerodynamic analysis, this project demonstrated the importance of a systematic approach to airfoil selection. By integrating theoretical derivations with computational analysis, a clear and justified decision was achieved. The methodology used in this study can be applied to other UAV design problems, providing a structured framework for evaluating aerodynamic performance.

Although the results are based on computational analysis using XFLR5, they provide

valuable insights into airfoil behavior. Future work may include advanced Computational Fluid Dynamics (CFD) simulations, wind tunnel testing, and full UAV system integration to further validate and refine the design.

In conclusion, the SD7037 airfoil is selected as the optimal choice for the Aurora UAV mission due to its superior aerodynamic efficiency, stability, and suitability for low Reynolds number flight conditions. This selection ensures that the UAV can achieve long-endurance flight while maintaining the stability required for accurate Northern Lights observation.

Chapter 14

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Chapter 15

Appendix

15.1 Machine Learning Model Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score

# Define files and airfoil names
files = {
    'S1223.csv': 'S1223',
    'SG6043.csv': 'SG6043',
    'SD7037.csv': 'SD7037',
    'E387.csv': 'E387',
    'NACA4412.csv': 'NACA4412'
}

all_data = []

print("--- Loading and Cleaning Data ---")
```



```
for file, name in files.items():
    try:
        temp_df = pd.read_csv(file, skiprows=11, sep=r'\s+', header=None)

        num_cols = temp_df.shape[1]
        standard_names = ['alpha', 'Cl', 'Cd', 'Cdp', 'Cm']

        if num_cols >= 5:
            extra_names = [f'Extra_{i}' for i in range(num_cols - 5)]
            temp_df.columns = standard_names + extra_names
        else:
            print(f"Warning: {file} has fewer than 5 columns.")
            continue

        temp_df['airfoil_name'] = name
        temp_df['Re'] = 365000

        all_data.append(temp_df)
        print(f"Loaded {name}")

    except FileNotFoundError:
        print(f"Error: {file} not found")

df_master = pd.concat(all_data, ignore_index=True)

# One-hot encoding
df_encoded = pd.get_dummies(df_master, columns=['airfoil_name'])

cols_to_drop = ['Cd', 'Cdp'] + [c for c in df_encoded.columns if 'Extra' in c]
X = df_encoded.drop(columns=cols_to_drop)
y = df_encoded['Cd']
```

```
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Models
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

rf_model = RandomForestRegressor(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

rf_pred = rf_model.predict(X_test)

print("Random Forest MAE:", mean_absolute_error(y_test, rf_pred))
print("Random Forest R2:", r2_score(y_test, rf_pred))

# Plot
plt.figure(figsize=(8,6))
plt.scatter(y_test, rf_pred, alpha=0.5)
plt.plot([y_test.min(), y_test.max()],
         [y_test.min(), y_test.max()],
         'r--')
plt.xlabel("Actual Cd")
plt.ylabel("Predicted Cd")
plt.title("ML Model Accuracy")
plt.grid(True)
plt.show()

# Ranking
ranking_results = []
```

```
for name in files.values():
    sample_input = pd.DataFrame(columns=X.columns)
    sample_input.loc[0] = 0

    sample_input['alpha'] = 2.0
    sample_input['Cl'] = 0.4
    sample_input['Re'] = 365000

    col_name = f'airfoil_name_{name}'
    if col_name in sample_input.columns:
        sample_input[col_name] = 1

    pred_cd = rf_model.predict(sample_input)[0]

    ranking_results.append({
        'Airfoil': name,
        'Predicted_Cd': pred_cd,
        'L/D': 0.4 / pred_cd
    })

rank_df = pd.DataFrame(ranking_results).sort_values(by='L/D', ascending=False)
print(rank_df)
```